20261013 Quiz

for (MATH301) Modern Algebra I

Jungwoo Kim
Department of Mathematics
Pohang University of Science and Technology

Exercise 1.7.18. Let H act on a set A. Define $a \sim b$ if $a = h \cdot b$ for some $h \in H$. Prove that this is an equivalence relation and that the equivalence classes (orbits) partition A.

Solution. For $a \in A$, $a = 1_H \cdot a$, proving reflexivity. If $a = h \cdot b$, then $b = h^{-1} \cdot a$, proving symmetry. If $a = h \cdot b$ and $b = k \cdot c$, then $a = (hk) \cdot c$, proving transitivity. Thus $\sim$ is an equivalence relation. Its class containing a is

$$[a] = \{h \cdot a : h \in H\} = H \cdot a,$$

exactly the orbit of a. Equivalence classes cover the set and are pairwise equal or disjoint, hence the distinct orbits partition A.

□

Exercise 2.2.4. For S_3, D_4, and Q_8, determine the centralizer of each element and the center of each group. Explain how Lagrange's theorem helps.

Solution. We compute $C_G(x) = \{g \in G : gx = xg\}$.

In S_3, a transposition commutes with itself and the identity, but not with either other transposition; the two nonidentity 3-cycles are powers of one another. Accordingly,

x	$C_{S_3}(x)$
1	S_3
$(12), (13), (23)$	$\langle x \rangle$
$(123), (132)$	$\langle (123) \rangle$

and $Z(S_3) = \{1\}$.

Write $D_4 = \langle r, s \mid r^4 = s^2 = 1,\ srs = r^{-1} \rangle$. We have $r^2 \in Z(D_4)$ and $r^i(sr^j) = (sr^j)r^i$ precisely when $r^{2i} = 1$. Two reflections sr^i, sr^j commute if and only if $r^{2(i-j)} = 1$, equivalently $i \equiv j \pmod 2$. Thus

x	$C_{D_4}(x)$
$1, r^2$	D_4
r, r^3	$\langle r \rangle$
s, sr^2	$\langle s, r^2 \rangle$
sr, sr^3	$\langle sr, r^2 \rangle$

and $Z(D_4) = \{1, r^2\}$.

In $Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$, the elements ± 1 are central. Distinct quaternion units among i, j, k anticommute, so an element $\pm i$ commutes precisely with $\langle i \rangle$ (and similarly for j, k):

x	$C_{Q_8}(x)$
± 1	Q_8
$\pm i$	$\langle i \rangle$
$\pm j$	$\langle j \rangle$
$\pm k$	$\langle k \rangle$

Therefore $Z(Q_8) = \{\pm 1\}$.

Lagrange's theorem restricts $|C_G(x)|$, since $\langle x \rangle \leq C_G(x) \leq G$. For example, a transposition in S_3 has order 2, so its centralizer has order 2 or 6; a single noncommuting element rules out 6. □

Exercise 2.2.6. Let $H \leq G$.

(a) Show $H \leq N_G(H)$; give a counterexample for a subset H which is not a subgroup.

(b) Show $H \leq C_G(H)$ if and only if H is abelian.

Solution.

(a) For $h \in H$, conjugation by h maps H to itself, and its inverse is conjugation by $h^{-1} \in H$. Hence $hHh^{-1} = H$, giving $H \leq N_G(H)$.
For a non-subgroup subset take $H = \{(12), (13)\} \subseteq S_3$. Conjugation by (12) sends (13) to (23), so $(12)H(12)^{-1} \neq H$. Thus $(12) \in H$ does not lie in $N_{S_3}(H)$.

(b) The inclusion $H \leq C_G(H)$ means exactly that every $h \in H$ commutes with every element of H, which is the definition that H is abelian.

□

Exercise 2.4.3. If H is an abelian subgroup of G, show that $\langle H, Z(G) \rangle$ is abelian. Give an example where $\langle H, C_G(H) \rangle$ is not abelian.

Solution. Every element of $Z(G)$ commutes with every element of G, and the elements of H commute with each other. Consequently, every element of $\langle H, Z(G) \rangle$ can be written hz with $h \in H$, $z \in Z(G)$. For two such elements,

$$(h_1 z_1)(h_2 z_2) = h_1 h_2 z_1 z_2 = h_2 h_1 z_2 z_1 = (h_2 z_2)(h_1 z_1),$$

proving this subgroup is abelian.

For an example let $G = D_4$ and $H = Z(D_4) = \langle r^2 \rangle$. Then H is abelian, while $C_G(H) = G$, so $\langle H, C_G(H) \rangle = G$ is nonabelian.

□

Exercise 2.4.7. Prove that $\langle (12), (13)(24) \rangle \leq S_4$ is isomorphic to D_4 (of order 8).

Solution. Set $a = (12)$, $b = (13)(24)$, $r = ab$, and $s = a$. Both a, b are involutions, and $r = ab$ is a 4-cycle, so $|r| = 4$. Moreover,

$$s^2 = 1, \qquad srs = a(ab)a = ba = (ab)^{-1} = r^{-1}.$$

The relations reduce every word in r, s to one of r^j or sr^j $(0 \leq j < 4)$, giving at most eight elements. The first four are distinct; the other four are distinct and do not coincide with them because s is a transposition while the elements of $\langle r \rangle$ are the identity, a double transposition, or

4-cycles. Thus there are exactly eight elements. Sending the standard generators of D_4 to r, s defines a surjective homomorphism between two groups of order 8, hence an isomorphism.

□

Exercise 2.4.14. A group is finitely generated if it has a finite generating set.

(a) Show every finite group is finitely generated.

(b) Show $\mathbb{Z}$ is finitely generated.

(c) Show every finitely generated subgroup of $(\mathbb{Q}, +)$ is cyclic.

(d) Deduce that $(\mathbb{Q}, +)$ is not finitely generated.

Hint. Clear the denominators of the proposed finite generating set.

Solution.

(a) A finite group H is generated by the finite set H itself.

(b) The additive group $\mathbb{Z} = \langle 1 \rangle$ is generated by one element.

(c) If $H = \langle a_1/b_1, \ldots, a_t/b_t \rangle \leq \mathbb{Q}$ with nonzero denominators $b_i > 0$, let $k = b_1 \cdots b_t$. Each $a_i/b_i = (a_i k/b_i)(1/k)$ is an integer multiple of $1/k$. Thus $H \leq \langle 1/k \rangle$, and every subgroup of a cyclic group is cyclic.

(d) Were $\mathbb{Q}$ finitely generated, (c) would make it cyclic. But if $\mathbb{Q} = \langle q \rangle$ with $q \neq 0$, then $q/2$ would have to be an integer multiple of q, forcing $1/2 \in \mathbb{Z}$. The generator $q = 0$ is plainly impossible. Hence $\mathbb{Q}$ is not finitely generated.

□

Exercise 2.4.15. Exhibit a proper subgroup of $(\mathbb{Q}, +)$ that is not cyclic.

Solution. Consider $H = \{m/2^k : m \in \mathbb{Z},\ k \geq 0\}$. It is closed under subtraction because fractions with denominators powers of 2 have a common denominator of this form. Thus $H \leq \mathbb{Q}$, but $1/3 \notin H$, so the subgroup is proper.

If $H = \langle q \rangle$ for a nonzero $q = m/2^k$, then $1/2^{k+1} \in H$ would have to equal tq for some $t \in \mathbb{Z}$. This would give $2tm = 1$, which is impossible. The zero element cannot generate H either. Thus H is not cyclic.

□

Exercise 2.5.3. Show that $\langle s, r^2 \rangle \leq D_4$ is isomorphic to V_4.

Solution. Since r^2 and s are commuting, distinct involutions, they generate exactly

$$H = \{1, r^2, s, sr^2\}.$$

The map $(\mathbb{Z}/2\mathbb{Z})^2 \to H$, $([i]_2, [j]_2) \mapsto r^{2i}s^j$, respects addition because the generators commute, and is a bijection between two four-element sets. Therefore $H \cong (\mathbb{Z}/2\mathbb{Z})^2 \cong V_4$.

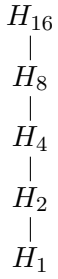
□

Exercise 2.5.9. Draw the subgroup lattices of $\mathbb{Z}/16\mathbb{Z}$, $\mathbb{Z}/24\mathbb{Z}$, and $\mathbb{Z}/48\mathbb{Z}$.

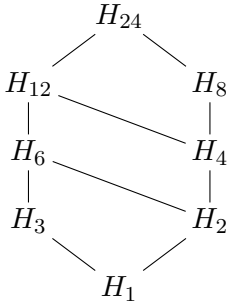
Hint. For cyclic groups there is a unique subgroup for each divisor of the group order.

Solution. In a cyclic group $\mathbb{Z}/n\mathbb{Z}$, for each divisor $d \mid n$ there is exactly one subgroup $H_d = \langle [n/d]_n \rangle$ of order d. Moreover, $H_d \leq H_e$ if and only if $d \mid e$. Therefore each subgroup lattice is the divisor lattice of n. The following Hasse diagrams connect exactly the covering relations:

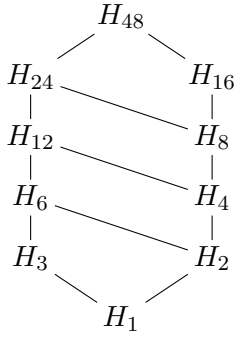
(a) $n = 16$:



(b) $n = 24$:



(c) $n = 48$:



□

Exercise 3.1.1. If $\varphi : G \to H$ is a homomorphism and $E \leq H$, show $\varphi^{-1}(E) \leq G$. If $E \trianglelefteq H$, prove $\varphi^{-1}(E) \trianglelefteq G$. Deduce $\ker \varphi \trianglelefteq G$.

Solution. Because $\varphi(1_G) = 1_H \in E$, the preimage is nonempty. If $x, y \in \varphi^{-1}(E)$, then $\varphi(xy^{-1}) = \varphi(x)\varphi(y)^{-1} \in E$, so $xy^{-1} \in \varphi^{-1}(E)$. Hence it is a subgroup.

If $E \trianglelefteq H$ and $g \in G$, $x \in \varphi^{-1}(E)$, then

$$\varphi(gxg^{-1}) = \varphi(g)\varphi(x)\varphi(g)^{-1} \in E.$$

Thus conjugation by g preserves $\varphi^{-1}(E)$ (equality follows by using g^{-1}), proving normality. Apply this to $E = \{1_H\}$ to obtain $\ker \varphi \trianglelefteq G$.

□

Exercise 3.1.2. Let $\varphi : G \to H$ be a homomorphism with kernel K. For $a, b \in \varphi(G)$ let $X = \varphi^{-1}(a)$ and $Y = \varphi^{-1}(b)$ be the fibers. If $u \in X$, $XY = Z$ in G/K, and $w \in Z$, prove that $w = uy$ for some $y \in Y$.

Hint. Consider $y = u^{-1}w$ and compute its image under φ.

Solution. The fibers are precisely the cosets of $K = \ker \varphi$. Since $u \in X$, the coset X equals uK. The equality $Z = XY$ implies

$$\varphi(w) = \varphi(u)\, b = ab.$$

Set $y = u^{-1}w$. Then

$$\varphi(y) = \varphi(u)^{-1}\varphi(w) = a^{-1}(ab) = b,$$

so $y \in \varphi^{-1}(b) = Y$, and $uy = w$ as required. This directly identifies the element of the second fiber rather than merely showing its existence.

□

Exercise 3.1.7. Define $\pi : (\mathbb{R}^2, +) \to (\mathbb{R}, +)$ by $\pi(x, y) = x + y$. Show it is a surjective homomorphism and describe its kernel and fibers geometrically.

Solution. For vectors $(x_1, y_1), (x_2, y_2)$, their images satisfy

$$\pi((x_1, y_1) + (x_2, y_2)) = x_1 + x_2 + y_1 + y_2 = \pi(x_1, y_1) + \pi(x_2, y_2).$$

The map is surjective because $\pi(c, 0) = c$ for every real c. Its kernel is

$$\ker \pi = \{(t, -t) : t \in \mathbb{R}\},$$

the line $x + y = 0$ through the origin of slope -1. For any $c \in \mathbb{R}$, the fiber $\pi^{-1}(c) = \{(x, y) : x + y = c\}$ is the parallel line $y = -x + c$, which is a coset of the kernel.

□